

Harry Desmond

CONTACT INFORMATION	Institute for Cosmology and Gravitation University of Portsmouth Dennis Sciamia Building Portsmouth PO1 3FX United Kingdom	<i>Email:</i> harry.desmond@port.ac.uk <i>Web:</i> https://harrydesmond.github.io <i>Citizenship:</i> Dual UK/USA <i>ORCID:</i> 0000-0003-0685-9791
RESEARCH INTERESTS	Astrophysical tests of fundamental physics, galaxy dynamics & phenomenology: modified gravity, dark matter, galaxy scaling relations, the galaxy–halo connection, local large-scale structure, (constrained) cosmological N-body and hydrodynamical simulations, tests of the cosmological principle Astrostatistics and machine learning: Field-level inference, symbolic regression, explainable AI, causal discovery Stellar structure and evolution	
EMPLOYMENT	Proleptic Senior Lecturer Institute for Cosmology and Gravitation, University of Portsmouth	01/2025 –
	Royal Society University Research Fellow Institute for Cosmology and Gravitation, University of Portsmouth	01/2022 – 12/2029
	McWilliams Fellow McWilliams Center for Cosmology, Carnegie Mellon University	10/2021 – 12/2021
	Junior Research Fellow University of Oxford (St John’s College)	10/2017 – 09/2021
EDUCATION	Stanford University PhD Physics <i>Thesis:</i> Dynamical constraints on the galaxy–halo connection <i>Supervisor:</i> Risa Wechsler	09/2012 – 06/2017
	University of Oxford (St John’s College) MPhys Astrophysics & Theoretical Physics, first class honours <i>Thesis:</i> The baryonic Tully–Fisher relation predicted by cold dark matter cosmogony <i>Supervisor:</i> Subir Sarkar	10/2008 – 06/2012
SELECTED AWARDS AND HONOURS	Royal Society University Research Fellowship (£1.76M total)	01/2022–12/2029
	Beecroft Visiting Fellowship (University of Oxford)	01/2026–05/2026
	Royal Society Enhanced Research Expenses awards	2021, 2023
	McWilliams Center Seed Grant (Carnegie Mellon University)	2021–2022

Balzan Fellowship, Centre for Cosmological Studies, New College (University of Oxford)	2017
Paul H. Kirkpatrick award for excellence in teaching (Stanford University)	2014
Gregory and Mary Chabolla Graduate Fellowship (Stanford University)	2012
Scott Prize for best overall performance in the MPhys examination (University of Oxford)	2012
BP Prize for best MPhys project in theoretical physics (University of Oxford)	2012
Winton Capital Prize for highest marks in the second-year Physics examinations (University of Oxford)	2010
Casberd Scholarship at St John's College (University of Oxford)	2009
39 th International Physics Olympiad in Hanoi, Bronze Medal	2008

SEMINARS, COLLOQUIA AND PRESENTATIONS Over 70 presentations in total, including 40 invited departmental seminars and colloquia at 27 institutions in 9 countries, and talks at 17 international conferences and workshops.

Forward-modelling a two-rung distance ladder: A 1.8% constraint on H_0

<i>Joint Astrophysics and Theory Seminar</i> , Imperial College London	03/2026
<i>CosmoVerse Seminar</i> (recording)	01/2026

Exhaustive Symbolic Regression: Learning (Astro)Physics directly from Data

<i>Symbolic Regression and Equation Discovery Workshop @ WCCI/CEC</i> , Maastricht, Netherlands (Invited)	06/2026
<i>Astrophysics Colloquium</i> , IAP (recording)	11/2025
<i>Cosmology Seminar</i> , Queen Mary University	10/2025
<i>Symbolic Regression in the Physical Sciences</i> , Royal Society workshop	04/2025
<i>High Energy Physics Seminar</i> , University of Southampton	11/2024
<i>PHYSTAT conference</i> , Imperial College London (poster)	09/2024
<i>COSMO21 conference</i> , Chania, Crete	05/2024
<i>Astrophysics Colloquium</i> , University of Oxford	02/2024
<i>Astroseminar</i> , University of Waterloo	01/2024
<i>Webinar on Mathematics, Physics and Machine Learning</i> , IST Lisbon (online here)	05/2023
<i>ICG Monday Lunch Talk</i> , University of Portsmouth	05/2023
<i>Cosmology Seminar</i> , University of Sussex	04/2023
<i>Cosmology Seminar</i> , Imperial College London	03/2023
<i>Cosmology Seminar</i> , University College London	01/2023

Symbolic Regression: Philosophy, Methods & Practice

<i>Cosmology Seminar</i> , University College London	10/2024
<i>Machine Learning Seminar</i> , Microsoft London	07/2024

Modified Newtonian Dynamics: Observational successes and challenges

<i>Rencontres de Moriond, Gravitation</i> , La Thuile, Italy (Invited)	04/2025
<i>Cosmology Seminar</i> , University of Oxford	04/2025

Testing Λ CDM with local large-scale structure, galaxies, and stars

<i>Astrophysics Seminar</i> , Bath	11/2024
<i>Universe Seminar</i> , IAP	10/2024
<i>Relativity and Cosmology Seminar</i> , QMUL	11/2023
<i>Dalitz Seminar</i> , University of Oxford	05/2022
<i>JBCA Colloquium</i> , University of Manchester	04/2022
<i>Cosmology Seminar</i> , University of Pennsylvania	12/2021
<i>Astronomy Seminar</i> , Case Western Reserve University	12/2021
<i>Cosmology Seminar</i> , University of Hawaii	12/2021
<i>ICG Astrophysics Colloquium</i> , University of Portsmouth	11/2021

Inferring Causation from Correlation

<i>ICG Forum</i> , Portsmouth	06/2024
<i>Cosmology Journal Club</i> , University of Oxford	01/2024

Four ways of looking at the Radial Acceleration Relation

<i>Cosmology Seminar</i> , University of Strasbourg	10/2024
<i>Cosmology Seminar</i> , Indian Institute of Technology, Hyderabad	04/2024
<i>Cosmology Seminar</i> , CEICO - FZU, Prague	02/2024
<i>Astronomy Seminar</i> , University of St Andrews	02/2024

Fundamental physics from galaxies

<i>Astrophysics Colloquium</i> , Institute for Theoretical Astrophysics, Oslo	01/2020
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Astrophysical tests of gravity

<i>The Paris–Saclay AstroParticle Symposium 2019</i>	10/2019
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A local resolution of the Hubble tension: The impact of screened fifth forces on the cosmic distance ladder

<i>Standard cosmology at the threshold of change?</i> , Thessaloniki (Invited)	06/2024
<i>Cosmic Controversies</i> , Chicago	10/2019

Galaxy velocity & size correlations, the halo Tully–Fisher relation and tests of the External Field Effect

<i>Bonn gravitation workshop</i>	09/2019
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Fifth force searches in galaxies

<i>PPPAC Seminar</i> , Imperial College London	03/2021
<i>Progress on Old and New Themes in Cosmology</i> , Avignon, France	12/2020
<i>Dark Energy in a Dark Age</i> , APCTP, Pohang, South Korea	10/2020
<i>CRAG Seminar</i> , Sheffield	10/2020
<i>Cosmology Seminar</i> , Institute for Theoretical Astrophysics, Oslo	01/2020
<i>The Paris–Saclay AstroParticle Symposium 2019</i>	10/2019
<i>The Cosmic Web workshop</i> , Edinburgh	06/2019
<i>15th Patras Workshop on Axions, WIMPs and WISPs</i> , Freiburg	06/2019
<i>Cosmology Seminar</i> , University College London	06/2019

<i>Astrophysics Seminar</i> , Imperial College London	05/2019
<i>ICG Colloquium</i> , Portsmouth	03/2019
<i>Cosmology Seminar</i> , Durham ICC	02/2019
<i>LSS group meeting</i> , Edinburgh	02/2019
<i>Particle Theory Seminar</i> , Nottingham CAPT	02/2019
<i>IAP Universe Journal Club</i> , Paris	10/2018
<i>Cosmology group meeting</i> , University of Pennsylvania, Philadelphia	10/2018
<i>Cosmology Seminar</i> , CITA, Toronto (recording)	07/2018
<i>Aquila Consortium group meeting</i> , Stockholm	03/2018

Gravitational maps and tests of screening

<i>Aquila Consortium group meeting</i> , MPIA, Garching	11/2017
<i>Cosmology group meeting</i> , University of Pennsylvania, Philadelphia	09/2017

Dynamical constraints on the galaxy–halo connection

<i>The Physics of Galaxy Scaling Relations</i> , Queen’s University, Kingston (poster)	07/2018
<i>Lunchtime Talk</i> , ESO, Garching	11/2017
<i>Quantifying and Understanding the Galaxy–Halo Connection</i> , KITP, University of California Santa Barbara (poster)	05/2017
<i>Cosmology Seminar</i> , University of Sussex	08/2016
<i>Cosmology Seminar</i> , University College London	08/2016
<i>Galaxies Discussion Group</i> , Cambridge	08/2016
<i>FLAT Talk</i> , Durham	08/2016
<i>Institute for Astronomy Seminar</i> , Edinburgh	08/2016

Testing halo abundance matching with galaxy dynamics

<i>2016 Munich Joint Conference, Discs in Galaxies</i> , ESO, Garching (poster)	07/2016
<i>SnowPAC 2016, The Galaxy–Halo Connection</i> , Snowbird, Utah, USA	03/2016

The Tully–Fisher and mass–size relations from halo abundance matching

<i>CosmoClub Cosmology Seminar</i> , UCSC	11/2015
<i>Santa Cruz Galaxy Workshop</i> , UCSC (recording)	08/2015
<i>KIPAC Tea</i> , Stanford	06/2015

RESEARCH SUPERVISION

29 students and 2 postdoctoral researchers, with around 40 papers led by my students and 45 by my supervisees. The 31 student entries below comprise 10 PhD or DPhil students — 5 of whom I supervised as principal thesis supervisor, the rest on side projects alongside their theses — and 21 master’s or undergraduate projects; two students appear in both lists, having returned to me for their PhDs. Supervisees listed without a co-supervisor were supervised solely by me.

Postdoctoral researchers:

<i>Indranil Banik</i>	08/2024 –
Local solutions to the Hubble and bulk flow tensions	

PhD students — principal thesis supervisor:

Antonio Nájera (PhD, Portsmouth) 2024 –
New handles on the cosmic distance ladder

Alicia Martin (DPhil, Oxford) 2023 –
Machine-learning fundamental physics from cosmology (w/ P. Ferreira, T. Yasin & D. Bartlett)

Richard Stiskalek (DPhil, Oxford) 2022 – 2026
Leveraging constrained large-scale structure simulations for new tests of physics (w/ J. Devriendt & A. Slyz)

Tariq Yasin (DPhil, Oxford) 2020 – 2023
Combining kinematic and photometric constraints on the galaxy–halo connection (w/ J. Devriendt & A. Slyz)

Deaglan Bartlett (DPhil, Oxford) 2019 – 2022
Fundamental physics from galaxies (w/ P. Ferreira)

PhD students — side projects:

Lukas Kammerer (PhD, Hagenberg) 2023 – 2026
Symbolic emulators for cosmological power spectra (w/ D. Bartlett & P. Ferreira)

Ekaterine Dadiani (PhD, CMU) 2021 – 2022
The galaxy–halo connection in the MassiveBlack-II simulation

Øyvind Christiansen (PhD, Oslo) 2021 – 2022
Simulating asymmetric correlation functions in modified gravity (w/ D. Mota)

Darsh Kodwani (DPhil, Oxford) 2018 – 2019
Screened modified gravity in relativistic perturbation theory

Kris Pardo (PhD, Princeton) 2018 – 2019
Testing self-interacting dark matter with galaxy warps (w/ P. Ferreira)

Masters and undergraduate project students:

Noah Pierce (BSc, Southampton) 2025
Testing the local supervoid with galaxy counts (w/ T. Yasin, R. Stiskalek & I. Banik)

Thenu Kaluarachchige (BSc, St Andrews) 2025
Inferring the Universe’s age from the Milky Way’s oldest stars (w/ I. Banik & R. Stiskalek)

Joshua Darne (MPhys, Oxford) 2025
The radial acceleration relation in hydrodynamical simulations (w/ T. Yasin & R. Stiskalek)

Aditya Naik (BSc, QMUL) 2024
Inferring causation from correlation

Catherine Gallagher (MPhys, Oxford) 2024 – 2025
The environment-dependence of galaxy properties (w/ T. Yasin & M. Jarvis)

Fedir Boreiko (BSc, Manchester) 2024 – 2025
Kinematic constraints on halo abundance matching (w/ T. Yasin, R. Stiskalek & M.

Jarvis)		
<i>Enoch Ko</i> (BSc, Warwick)		2024 – 2025
Quantifying Renzo’s rule (w/ T. Yasin, R. Stiskalek & M. Jarvis)		
<i>James Harvey</i> (BSc, Oxford)		2024
Predicting star formation histories from halo mergers (w/ T. Yasin & R. Stiskalek)		
<i>Param Upadhyay</i> (BSc, Tufts)		2023
A second look at the bulk flow of CosmicFlows-4 (w/ S. von Hausegger & S. Sarkar)		
<i>Amelia Ford</i> (BSc, Southampton)		2023 – 2024
Symbolically regressing luminosity and mass functions (w/ D. Bartlett & P. Ferreira)		
<i>Dhruva Patel</i> (BSc, Open University)		2023 – 2024
Testing the isotropy of the universe		
<i>Tomas Sousa</i> (MPhys, Nottingham)		2023
Machine-learning the inflaton potential (w/ P. Ferreira & D. Bartlett)		
<i>Nazanin Azimi</i> (BSc, CMU)		2022
Emulating stellar structure simulations		
<i>Maxwell Hutt</i> (MPhys, Oxford)		2020 – 2022
Topics in constrained N-body simulation (w/ J. Devriendt & A. Slyz)		
<i>Alicia Martin</i> (MPhys, Manchester)		2020
Symbolically regressing halo density profiles (w/ P. Ferreira, T. Yasin & D. Bartlett)		
<i>Richard Stiskalek</i> (BSc, Glasgow)		2019
Halo abundance matching with total baryonic mass		
<i>Isabel Sands</i> (BSc, Yale)		2019
Numerical solutions for scalar field profiles (w/ P. Ferreira)		
<i>Robert Clemenson</i> (MPhys, Oxford)		2019
Non-gaussianity of weak lensing likelihoods (w/ S. Joudaki)		
<i>Thomas Holvey</i> (MPhys, Oxford)		2018 – 2019
Baryon mass functions from ALFALFA $\times$ SDSS		
<i>Eliza Dickie</i> (MPhys, Oxford)		2018 – 2019
Fifth-force observables in hydrodynamical simulations		
<i>William Martin</i> (BSc, Cambridge)		2018
Dynamical probes of screened fifth forces		

TEACHING	Lecturer: Complex analysis & integral transforms (Portsmouth)	2023 –
	Course designer & lecturer: <i>Exploring galaxies with the Sloan Digital Sky Survey</i> (CMU)	2022
	Tutor: Cosmology 4 th year MMathPhys course (Oxford)	Hilary 2021
	Tutor: Astrophysics 4 th year MPhys major option (Oxford)	2020 – 2021
	Tutor: Mechanics CP1 (Oxford)	Hilary 2018
Stanford Teaching Assistantships		
	Cosmology (<i>Phys 361</i> ; graduate level)	Winter 2016

	Intermediate Electromagnetism II (<i>Phys 121</i> ; 2nd yr undergraduate) Spring 2015	
	Introduction to Laboratory Physics (<i>Phys 67</i> ; 1st yr undergraduate) Spring 2014	
	Mechanics and Special Relativity (<i>Phys 61</i> ; 1st yr undergraduate) Autumn 2013	
	Modern Physics Lab (<i>Phys 26</i> ; physics for non-majors) Spring 2013	
	Physics TA mentor (Stanford Mentors in Teaching Program) 2014 – 2017	
	Mentor, “Gravity! From the Big Bang to Black Holes” (MOOC run by Paris Diderot) 2016	
	Stanford Splash! 2014 – 2017	
	<i>Taught 8 standalone classes on philosophy and literature</i>	
LANGUAGES	Python, JAX, C/C++, Mathematica, Fortran, Matlab, SQL, ROOT	
	Simulation codes: RAMSES, MESA	
	English (native), French (intermediate), German (basic)	
SERVICE	Referee for <i>MNRAS</i> , <i>ApJ</i> , <i>JCAP</i> , <i>A&A</i> , <i>MDPI (Galaxies, Particles)</i> , <i>IJMPA</i> , <i>IJMPD</i> , <i>EPJP</i> , <i>MPLA</i> , <i>Phys. Lett. B</i> , <i>Advances in Astronomy</i> , <i>Ap&SS</i> , <i>Open J. of Astrophysics</i> , <i>Int. J. Theor. Phys.</i> , <i>Class. Quantum Grav.</i> , <i>Nuclear Physics B</i> , <i>PRD</i> , <i>PRR</i> , <i>PRL</i> , <i>Science Bulletin</i> , <i>PASA</i> , <i>JHEAp</i> , <i>Phil. Trans. R. Soc. A</i> , <i>Results in Physics</i> , <i>Scientific Reports</i> and <i>Science</i> 2016 –	
	External PhD thesis examiner (Nottingham, Durham) 2025 –	
	External grant reviewer for the University of Malta 08/2026	
	Referee for the Symbolic Regression and Equation Discovery Workshop @ WCCI/CEC, Maastricht, Netherlands 06/2026	
	Member of the Aquila Consortium Editorial Advisory Group 2022–2023 & 2025–2027	
	Guest Editor of Special Issue “Symbolic Regression in the Physical Sciences” (<i>Philosophical Transactions of the Royal Society A</i> ; from the Royal Society meeting of the same name) 2026	
	Guest Editor of Special Issue “The Hubble Tension in Cosmology” (<i>Galaxies</i>) 2025	
	STFC Small Award reviewer 2024	
	Member of the MIGHTEE collaboration 2023 –	
	PhD progression panellist (Portsmouth; 5 students to date) 2023 –	
	ICG Senior Management Committee (Portsmouth) 2022 –	
	ICG Colloquium organiser (Portsmouth) 2022 –	
	Internal fellowship assessor (Portsmouth) 2022	
	EuCAPT Portsmouth & Sussex rep. and Training task force member 2022 –	
	McWilliams Journal Club organiser (CMU) 2021 – 2022	
	External grant reviewer for the Polish National Science Centre (×2) 2020 – 2021	
	MMathPhys master’s project assessor (Oxford) 2019	
	Academic Services Committee (St John’s College) 2019 – 2020	

	Cosmology Reading Group organiser (Oxford)	2017 – 2019
	Novel Probes Project, co-founder; a lead and corresponding author of its two <i>Reviews of Modern Physics</i> reviews	2017
WORKSHOP ORGANISATION	Fundamental Physics from Galaxies one-day workshop (Oxford)	09/2026
	National Astronomy Meeting parallel session “ <i>Theoretical and observational approaches to the Hubble tension</i> ” (Durham)	07/2025
	Symbolic Regression in the Physical Sciences (Royal Society, London)	04/2025
	<i>Stellar Tests of Gravity</i> workshop (CMU)	03/2022
	Aquila Consortium group meeting (St John’s College, Oxford)	05/2019
OUTREACH	Interviewed for <i>New Scientist</i> (article)	08/2024
	Interviewed for <i>The Sensemaker</i> (article)	05/2024
	Wrote article for <i>The Conversation</i> <i>165,108 reads; also published in Spanish translation</i>	05/2024
	Interviewed for <i>New Scientist</i> (article)	03/2024
	SciTech Day career panelist (Carnegie Science Center, Pittsburgh)	12/2021
	Oxford Stargazing, Cosmology stall	2018 – 2019
	St John’s College	
	Inspire Digital column “ Dr Harry’s Questions ”	2019 – 2020
	Research Blast	02/2020
	Teachers’ Residential Programme mini-seminar	04/2019
	Publicity officer for Oxford University Space and Astronomical Society	2010 – 2011
TRAINING	Introduction to Management (Royal Society)	07/2023
	Leadership Effectiveness (Royal Society)	10/2023
PUBLICATIONS	arXiv	
	NASA ADS	
	INSPIRE HEP	